# DSS : Duivestein Storage System

## DSS Software Interface

This document describes the standard software interfaces between DSS and external programs, such as cutting optimizers, nesting software and ERP-systems. The appendix contains several examples of the imports and exports discussed in the paragraphs below.

## IMPORTS

DSSServer uses a built-in scheduler to read import files from a local folder, network share or from an ftp-server. The scheduler runs the imports at regular intervals. The next paragraphs describe the standard imports that are available in the software. Custom or user defined imports are not listed. The DSS software is regularly extended with new imports to facilitate communication with new machines, external software or systems.

### Production order

A production order, in DSS called a Production batch, is a list of one or more articles together with a list of one or more route steps. These can be supplied as an xml-file where each file results in one production batch, or as a csv-file where all the lines in the file will be converted to a group of production batches.

#### *Batch-xml*

The batch-xml is a structured format for simple production orders (e.g. “move three sheets to a panel saw” or “put one sheet on a nesting machine”) as well as more complex instructions (e.g. “move seven sheets in stacks of two to the label station, start the label robot after each sheet, move the sheets to a panel saw in stacks of two and print a report when the last sheet has been put down on the saw”). Batch-xmls can also be used for simple order picking. [➔ *Batch-xml example*]

A batch consists of one or more Plans (e.g. saw layout, nesting schema, label plan, order line, …) and one or more Routes (steps the sheets have to take to get from A to B). In the simplest form, a batch-xml contains only an identifier for the batch, a list of sheets to handle, and the name of the machine or output bin to store the sheets on.

#### *Batch-lst*

The batch-lst is a tab-separated csv-file where each line refers to one or more identical sheets with their respective crane movements. Each line will result in one production batch in DSS, and all batches originating from the same file and date/time of import will be grouped together. This way, the operator can start a whole list of production batches with just one click in DSS. [➔ *Batch-lst example*]

### *CTS-file*

DSS can read several CTS (*Cut To Saw*, optimizer) file types directly, such as Pattern Exchange/PTX-files, Optiplanning XML-files, bNest BWKL-files and Mayer MAY-files. It translates these to production batches. [➔ *CTS example*]

## Picking order

DSS supports two imports for picking orders. The first format is specific for picking orders, the second a subset of the batch-xml.

### *Picklist-xml*

The picklist-xml is used for regular order picking. The xml contains a list of articles and quantities and the name of the station where the collected sheets have to be put. [➔ *Picklist-xml example*]

### *Picking-xml (batch)*

The batch-xml-format is used when the picking process consists of more than one step and/or requires extra postprocessing, such a signaling a packing machine. [➔ *Picking-xml (batch) example*]

## Recipe

With a recipe one can control the production of multiple sheets over several stations, allowing for raw material to be combined to semi-finished products and much more. Recipes are stored as xml-files. [➔ *Recipe-xml example*]

## Label

Label data can be provided as either an xml-file or a csv-file. In both cases the minimum information required are the coordinates of the label relative to the sheet and some data to print on the labels. The file describes for each label the location on the sheet where the label has to be put, optionally rotated, and what data to print on the label. This label data can be a list of alphanumeric fields, a reference to a bitmap file or a combination of both.

Each sheet on the label station requires one unique label file containing all the label data for that sheet. If a sheet is to be put on the label station but no labels should be printed, an “empty” label file has to be provided or that sheet.

### *Label-csv*

The label-csv contains only the data described above. The layout of the label is defined in a separate template file that refers to columns of the csv-file for the actual label-by-label information. During printing, the template and label-csv are automatically merged and the resulting label is sent to the printer. [➔ *Label-csv example*]

### *Label-xml*

The label-xml contains both the label data and the label layout. [➔ *Label-xml example*]

## Materials and articles

t.b.d.

## Crane jobs and loading sheets

Imports can be used to create (semi-)automated crane jobs. For each (set of) sheet(s) the details of the sheet(s) have to be provided.

### *Cranejob-xml*

To start a crane job directly and automatically, DSS uses the data from a cranejob-xml. Besides the sheet details, this xml also contains information about where to pick up the sheet and/or on what stackposition to put it down again. [➔ *Cranejob-xml example*]

### *Deposit-xml*

t.b.d. [➔ *Deposit-xml example*]

### *Loadingdocklist-xml*

t.b.d. [➔ *Loadingdocklist-xml example*]

### *Restpiecelist-xml*

Rest pieces are stored automatically or semi-automatically, based on a setting in the DSS software. For semi-automatic mode, the operator is presented with the pop-up dialog in which sheet-details are shown and he has to confirm or reject loading of the sheet(s) – either sheet-by-sheet, or for the entire list. [➔ *Restpiecelist-xml example*]

## EXPORTS

DSS features a wide variety of exports in txt, csv, xml, html or proprietary file formats. Besides these standard exports, one can generate files from any custom SQL-SELECT-statement or even through scripting.

By default exports are stored in a local folder or network share. After being generated the export can also automatically be printed, sent to an ftp-server, emailed to one or a list of recipients and/or be sent by sms.

DSSServer uses a built-in scheduler to generate the exports. The scheduler runs at regular intervals or on specific times of the day. Each export can be limited to specific days of the week, or run only in defined time slots.

The next paragraphs describe the standard exports that are available in the software. Custom or user defined exports are not listed. The DSS software is regularly extended with new exports to facilitate communication with new machines, external software or systems.

### Settings and options

Parameters for exports are defined in the main ini-file for DSSServer. It contains a list of all exports used for the current configuration, and details for each export in a separate section. For example:

```
[FILEEXPORTS]
DSSStockSummaryExport=STOCKEXPORT_HTML

[STOCKEXPORT_HTML]
FOLDER=\\SERVER\SHARE\DSS-Exports\
FILENAME=CURRENTSTOCK.HTML
INCLUDERESTSHEETS=True
TIMINGSEQ=15M
```

The current stock is stored as `\\SERVER\SHARE\DSS-Exports\CURRENTSTOCK.HTML` every 15 minutes.

### *Scheduler options*

The scheduler is triggered by the main event loop of DSSServer, which usually occurs every 10 seconds. On every occurrence, the timings parameters and optional extra rules are checked. If all conditions are met, the export is actually executed.

### *Timing parameters*

- RUNAT : Semi-colon separated list of points of time when to run the export.
- RUNBETWEEN : Semi-colon separated list of time windows when the export is allowed to run.
    - Example : 08:00:00-12:15:00;13:15-18:00
- RUNON : Semi-colon separated list of days of the week and/or days of the month on which to run the export.
    - Example : MON;TUE;WED;THU;FRI
    - Example : 1;15
- TIMINGSEQ : A list of number of days (D), hours (H), minutes (M) and/or seconds (S) between two runs of the export.
    - Example : 6H → Run every 6 hours
    - Example : 15M → Run every 15 minutes
- CHECKMISSED : Run export on the first moment *after* the RUNON/RUNAT-moment has passed. Choices = True, False. Default = False.

### *Extra rules*

- SQLCONDITION : SQL-select statement the must result in a value bigger than zero for the export to run.
    - Example : `SELECT COUNT(*) FROM CRANEMOVEMENT WHERE (ENTRYDATE < DATEADD(-5 MINUTE TO CURRENT_TIMESTAMP))` → Run if there is any CRANEMOVEMENT-record present where ENTRYDATE is older than 5 minutes ago.
- SENSORCONDITION : Rule to check any of the machine sensors that are defined in DSS for the export to run. A condition consists of a sensor name, a comparator (“=”, meaning “equal”; or “#”, meaning “not equal”) and a value (EMPTY, BUSY, WORKING, DONE, ERROR, AUTO, STOP, MANUAL) to compare the sensor status with. Each export can contain more than one SENSORCONDITION. An export will run if *all* SENSORCONDITIONs are met.
    - Example : $KRF#BUSY → Run if sensor status “$KRF” is not equal to “BUSY”
    - Example : MDS=MANUAL → Run if sensor status “MDS” is equal to “MANUAL”

### *Formatting options*

The format options below apply to most, but not all, of the exports in DSS.

- EXPORTFORMAT : File format. Choices = TEXT, CSV, HTML, XML. Default = <derived from filename>.
- COLUMNSEPARATOR : Character or ascii-value of a character to separate data in text- and csv-files. Default = 9 (meaning: tab-character).
- XMLROOTNAME : Name of the root-node for xml-files. Default = “<export type>”.

- XMLITEMNAME : Name of the item-node for xml-files. Default = “ITEM”.
- HTMLREFRESHINTERVAL : Interval (in seconds) to automatically reload an html-file. A value of -1 will set the interval to the same value as the TIMINGSEQ, or 120 seconds if TIMINGSEQ is not defined. Default = 0 (meaning: do not reload).

### *Print options*

Many of the print options can be set in de main ini-file as well as in the export section. The values taken from the main ini-file are used as defaults that are overridden by values from the export section.

- PRINT : To print or not to print, that’s the setting. Choices: True, False. Default = False.
- PRINTERNAME : Name of the printer, as can be seen in the Windows print manager. Alternative : ini-file, section “PRINTER”, item “NAME”.
- PRINTERMARGINLEFT : Left margin (in mm). Alternative : ini-file, section “PRINTER”, item “MARGINLEFT”. Default = 10.
- PRINTERMARGINTOP : Top margin (in mm). Alternative : ini-file, section “PRINTER”, item “MARGINTOP”. Default = 10.
- PRINTERMARGINRIGHT : Right margin (in mm). Alternative : ini-file, section “PRINTER”, item “MARGINRIGHT”. Default = 10.
- PRINTERMARGINBOTTOM : Bottom margin (in mm). Alternative : ini-file, section “PRINTER”, item “MARGINBOTTOM”. Default = 10.
- PRINTERSCALE : Scale factor for printing (in %). Alternative : ini-file, section “PRINTER”, item “SCALE”. Default = 100.
- PRINTERORIENTATION : Page orientation. Choices = PORTRAIT, LANDSCAPE. Alternative : ini-file, section “PRINTER”, item “ORIENTATION”. Default = PORTRAIT.

### *FTP-options*

- FTPHOST : IP-address or hostname of the ftp-server.
- FTPPORT : TCP-port for connecting to the ftp-server. Default = 21.
- FTPUSERNAME : User name to log on to the ftp-server. Default = anonymous.
- FTPPASSWORD : Password to log on to the ftp-server. The password shown here is automatically scrambled by DSSServer.
- FTPUSETLS : Use Transport Layer Security. Choices = NOTLS, IMPLICITTLS, REQUIRETLS, EXPLICITTLS. Default = NOTLS.
- FTPREMOTEDIR : Remote directory on the ftp-server. Default = / (meaning: root).
- FTPDELETELOCAL : Delete local file after uploading. Choices = True, False. Default = False.

### *Email options*

Sending email is done through an smtp-server.

- EMAILSUBJECT : Title of the report. Default = “DSS Report : <export type>”.
- EMAILTO : Email address or semi-colon separated list of email dresses to send the report to.
- EMAILCC : Email address or semi-colon separated list of email dresses to send a carbon copy to.
- EMAILBCC : Email address or semi-colon separated list of email dresses to send a blind carbon copy to.

- EMAILFROM : Email address of the sender.
- EMAILHOST : IP-address or host name of the mail server.
- EMAILPORT : TCP-port for connecting to the mail server. Default = 25.
- EMAILUSETLS : Use Transport Layer Security. Choices = NOTLS, IMPLICITTLS, REQUIRETLS, EXPLICITTLS. Default = NOTLS.
- EMAILAUTHENTICATION : Use authentication. Choices = NONE, DEFAULT, SASL. Default = DEFAULT.
- EMAILUSERNAME : User name to log on to the mail server.
- EMAILPASSWORD : Password to log on to the mail server.
- EMAILBODY : One line or short note to describe the report. Default = “The report can be found attached to this email.”.
- EMAILATTACHMENTASBODY

### *SMS-options*

- SMSFROM : Phone number of the sender.
- SMSTO : Phone number or semi-colon separated list of phone numbers of the recipient(s).

## Current stock exports

Stock exports list all (available) materials in the warehouse. These can be summarized to one line per material, or detailed up to one line per sheet. Some exports are tailored to specific external systems. Use export settings to define the format and contents of the exports.

### *DSSStockCheckExport*

This export lists the amount of sheets that can or have to be added to the stock.

Columns for MINIMUM: MATERIAL, COLOR, SUPPLIER, STATUS, DENSITY, XDIMSIZE, YDIMSIZE, ZDIMSIZE, GRAIN, ARTICLECODE, STOCK, MINIMUM, ADDSHEETS.

Columns for MAXIMUM: MATERIAL, COLOR, SUPPLIER, STATUS, DENSITY, XDIMSIZE, YDIMSIZE, ZDIMSIZE, GRAIN, ARTICLECODE, PACKAGESIZE, STACKPOSITION, STOCK, ADDSHEETS, ADDPACKAGES.

Options:

- CHECKTYPE : List the minimum number of sheets that have to be added to the warehouse to comply with the “minimum stock amount” setting for each material, or the maximum number of sheets that can be stored in the warehouse. Choices = MINIMUM, MAXIMUM. Default = MINIMUM.

### *DSSStockDetailExport*

This is a detailed list of all the sheets currently in stock. The list contains an entry for each and every sheet.

Columns: MATERIAL, MATERIALNAME, STACKPOSITION, ZINDEX, ROTATION, XDIMSIZE, YDIMSIZE, ZDIMSIZE, SHEETBARCODE, BARCODE, ARTICLECODE, WEIGHT, GRAIN, FILTER1, FILTER2, FILTER3, FILTER4, FILTER5, FILTER6, FILTER7, FILTER8, FILTER9.

Options:

- None

### *DSSStockSummaryExport*

This is a basic summarized list of all the sheets currently in stock. The list contains entries for each unique combination of material, length, width and grain. Using the Options described below, extra columns or lines can be included.

Standard set of columns: MATERIAL, COLOR, SUPPLIER, STATUS, HASGRAIN, DENSITY, ZDIMSIZE, GRAIN, BARCODE, ARTICLECODE, QUANTITY.

Alternative set of columns, using INCLUDEDIMENSIONS=True: MATERIAL, COLOR, SUPPLIER, STATUS, HASGRAIN, DENSITY, XDIMSIZE, YDIMSIZE, ZDIMSIZE, GRAIN, QUANTITY.

Options:

- INCLUDERESTSHEETS : Include rest sheets in the stock count, or only standard sized sheets. Choices : True, False. Default : True.
- INCLUDEDIMENSIONS : Include columns with XDIMSIZE and YDIMSIZE (meaning: length and width) of sheets. Choices : True, False. Default : <not INCLUDERESTSHEETS>.
- INCLUDEEMPTYLINES : Include lines for standard sized sheets without stock. Choices : True, False. Default : True.
- INCLUDEINOUTPOSITIONS : Include sheets currently stored in IN of IN/OUT positions. Choices : True, False. Default : True.

A set of examples is included in the appendix [➔ *Stocksummary examples*]

### *DSSMaterialsExport*

The data structure of “Materials” in DSS consists mainly of two tables:

- “MATERIAL” : The substance a sheet is made from, defined by a name, an optional color, the density and the thickness (ZDIMSIZE). Length and width are not properties of a material.
- “MATERIALSIZE” : The unique combination of a material, length (XDIMSIZE), width (YDIMSIZE) and grain, also referred to as an “Article”.

The DSSMaterialsExport generates a table (text, csv, html) or structured file (xml) from all materials and articles.

Columns: ID, EXTERNALID, NAME, COLOR, SUPPLIER, MATERIALSTATUS, HASGRAIN, DENSITY, ZDIMSIZE, SEPERATORTHICKNESS, BLOW, BEND, PUMP, HANDLING, PARAMETERS, ARTID, ARTEXTERNALID, XDIMSIZE, YDIMSIZE, GRAIN, BARCODE, ARTICLECODE.

Options:

- None

### *ArdisStockExport*

A tab-separated csv-file containing all the material properties relevant for the Ardis Cutting Optimizer software including the (free) stock quantity. The quantity is, depending on the value of SPLITRESERVED, calculated using total stock or free stock.

The “TRIM” and “MIN_” settings are available at the material-level – defined in the material parameters – with a default taken from the export options as listed below.

Standard set of columns: MATERIAL, GRAIN, LENGTH, WIDTH, THICKNESS, INFO1, INFO2, INFO3, INFO4, INFO5, INFO6, INFO7, INFO8, INFO9, LOCATION, QUANTITY, TRIM_LEFT, TRIM_RIGHT, TRIM_UPPER, TRIM_BOTTOM, REST_MIN_L, REST_MIN_W, REST_MIN_AREA, SUPPLIER.

Alternative set of columns, using SPLITRESERVED=True: MATERIAL, GRAIN, LENGTH, WIDTH, THICKNESS, INFO1, INFO2, INFO3, INFO4, INFO5, INFO6, INFO7, INFO8, INFO9, LOCATION, QUANTITY, RESERVED, TRIM_LEFT, TRIM_RIGHT, TRIM_UPPER, TRIM_BOTTOM, REST_MIN_L, REST_MIN_W, REST_MIN_AREA, SUPPLIER.

Options:

- SPLITRESERVED : Add a “RESERVED” column containing the number of sheets that are reserved in current production. Choices = True, False. Default = False.
- SUMMARIZE : Summarize quantities for each unique combination of material, dimensions, grain and filters. When summarized, the LOCATION column contains a dash (“-“) instead of the stackposition alias. Choices = True, False. Default = False.
- ALLMATERIALS : List also all articles without (free) stock. Choices = True, False. Default = False.
- TRIMLEFT : Trim on the left side of the sheet (in mm). Overridden by the material parameter “TRIMLEFT”.
- TRIMRIGHT : Trim on the right side of the sheet (in mm). Overridden by the material parameter “TRIMRIGHT”.
- TRIMUPPER : Trim on the upper side of the sheet (in mm). Overridden by the material parameter “TRIMUPPER”.
- TRIMBOTTOM : Trim on the bottom side of the sheet (in mm). Overridden by the material parameter “TRIMBOTTOM”.
- RESTMINLENGTH : Value for minimum rest piece length (in mm). Overridden by the material parameter “RESTMINLENGTH”.
- RESTMINWIDTH : Value for minimum rest piece width (in mm). Overridden by the material parameter “RESTMINWIDTH”.
- RESTMINAREA : Value for minimum rest piece area (in m2). Overridden by the material parameter “RESTMINAREA”.

### *bNestMaterialExport*

This export lists all the materials as required by bNest nesting software in a tab-separated csv-file.

Columns: MATERIAL, LENGTH, WIDTH, THICKNESS, ESSENCE, GRAIN, TRIMMING.

Options:

- TRIMMING = Trim value (in mm).

### *VentawinStockExport*

This export lists all the materials as required by Ventawin ERP software in a semi-colon-separated csv-file. The RESERVED column is fixed to “0”.

Columns (no column titles): ARTICLECODE, QUANTITY, MATERIAL, LENGTH, WIDTH, THICKNESS, RESERVED.

Options:

- None

## Stock movement exports

For each change in the stock, DSS can generate file for external systems to do an incremental synchronization or to track sheets used in production. DSS supports stock movement feedback as xml, csv and json-files, as well as custom scripts. The xml and csv-files contain all the details of the sheet that has just been moved, like material, dimensions, former and current stack position. The json-file is less detailed. [➔ *Stockmovement-xml example*]

## System status exports

These exports describe the current system status. They are used for signaling problems and they can assist in preventing them.

### *DSSCraneStatusExport*

The Crane Status Export lists the most recent machine alarms (errors, warnings and messages). This export is usually executed with an SQLCONDITION counting the number of cranemovements that are still in progress.

Columns: ENTRYDATE, MESSAGE, REMARKS.

Options:

- LINECOUNT : The number of lines in the report. Default = 20.

### *DSSConflictDetailsExport*

This export generates an extensive listing of all cranemovement issues, together with the machine alarms, movement parameters and information about performance.

Columns: Vary widely, depending on used options.

Options:

- PERIOD : Period for which to generate the report. Default = 1D.
- ADDALARMDESCRIPTIONS : Choices = True, False. Default = True.
- ADDPLCDESCRIPTIONS : Choices = True, False. Default = True.
- INCLUDEERRORS : Choices = True, False. Default = True.
- INCLUDEWARNINGS : Choices = True, False. Default = True.
- INCLUDEMESSAGES : Choices = True, False. Default = True.
- INCLUDEPLCDATA : Choices = True, False. Default = True.
- INCLUDEUNUSEDALARMS : Choices = True, False. Default = True.
- INCLUDEUNUSEDPLCDATA : Choices = True, False. Default = True.
- INCLUDEFAILED : Choices = True, False. Default = False.
- INCLUDEALARMSDURINGRESOLVE : Choices = True, False. Default = True.
- MARKFIRSTERROR : Choices = True, False. Default = False.

## Generic exports

These exports are highly flexible and not linked to one specific application within the system.

### *DSSGenericSQLExport*

Using the Generic SQL Export, DSS can store virtually any information that can be retrieved through an SQL-statement. [➔ *SQL export example*]

### *DSSScriptExport*

The Script Export allows scripting to generate any kind of file.

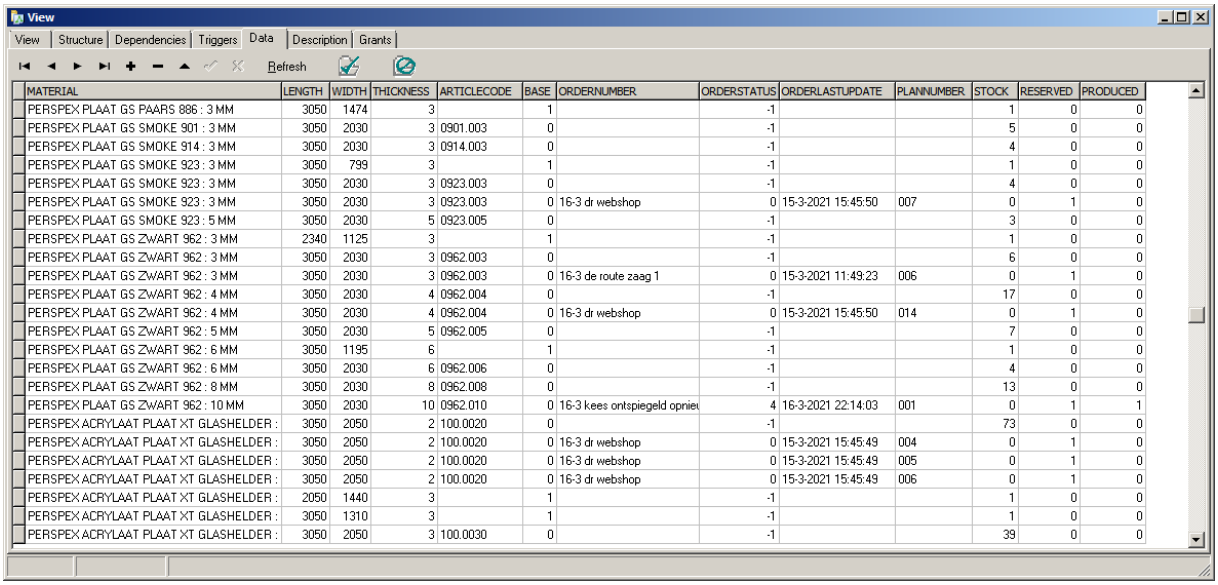
## REST-API

The DSSServer uses a REST-interface to communicate with DSSClient and DSS utilities. Part of this API is available to third party developers for interfacing with DSS, for example for requesting actual stock data and current jobs or batches.

For more information about the REST-API, please contact DSS Automation.

## DATABASE LINK

To circumvent the need for exporting and importing files, database views can be offered to third party developers, for example for retrieving stock information or details on stock reservations. Options on datasets and columns in these views are defined per customer. An example:




| MATERIAL | LENGTH | WIDTH | THICKNESS | ARTICLECODE | BASE | ORDERNUMBER | ORDERSTATUS | ORDERLASTUPDATE | PLANNUMBER | STOCK | RESERVED | PRODUCED |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PERSPEX PLAAT GS PAARS 886 : 3 MM | 3050 | 1474 | 3 | | 1 | | -1 | | | 1 | 0 | 0 |
| PERSPEX PLAAT GS SMOKE 901 : 3 MM | 3050 | 2030 | 3 | 0901.003 | 0 | | -1 | | | 5 | 0 | 0 |
| PERSPEX PLAAT GS SMOKE 914 : 3 MM | 3050 | 2030 | 3 | 0914.003 | 0 | | -1 | | | 4 | 0 | 0 |
| PERSPEX PLAAT GS SMOKE 923 : 3 MM | 3050 | 799 | 3 | | 1 | | -1 | | | 1 | 0 | 0 |
| PERSPEX PLAAT GS SMOKE 923 : 3 MM | 3050 | 2030 | 3 | 0923.003 | 0 | | -1 | | | 4 | 0 | 0 |
| PERSPEX PLAAT GS SMOKE 923 : 3 MM | 3050 | 2030 | 3 | 0923.003 | 0 | 16-3 dr webshop | 0 | 15-3-2021 15:45:50 | 007 | 0 | 1 | 0 |
| PERSPEX PLAAT GS SMOKE 923 : 5 MM | 3050 | 2030 | 5 | 0923.005 | 0 | | -1 | | | 3 | 0 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 3 MM | 2340 | 1125 | 3 | | 1 | | -1 | | | 1 | 0 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 3 MM | 3050 | 2030 | 3 | 0962.003 | 0 | | -1 | | | 6 | 0 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 3 MM | 3050 | 2030 | 3 | 0962.003 | 0 | 16-3 de route zaag 1 | 0 | 15-3-2021 11:49:23 | 006 | 0 | 1 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 4 MM | 3050 | 2030 | 4 | 0962.004 | 0 | | -1 | | | 17 | 0 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 4 MM | 3050 | 2030 | 4 | 0962.004 | 0 | 16-3 dr webshop | 0 | 15-3-2021 15:45:50 | 014 | 0 | 1 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 5 MM | 3050 | 2030 | 5 | 0962.005 | 0 | | -1 | | | 7 | 0 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 6 MM | 3050 | 1195 | 6 | | 1 | | -1 | | | 1 | 0 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 6 MM | 3050 | 2030 | 6 | 0962.006 | 0 | | -1 | | | 4 | 0 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 8 MM | 3050 | 2030 | 8 | 0962.008 | 0 | | -1 | | | 13 | 0 | 0 |
| PERSPEX PLAAT GS ZWART 962 : 10 MM | 3050 | 2030 | 10 | 0962.010 | 0 | 16-3 kees ontspiegeld opnie | 4 | 16-3-2021 22:14:03 | 001 | 0 | 1 | 1 |
| PERSPEX ACRYLAAT PLAAT XT GLASHELDER : | 3050 | 2050 | 2 | 100.0020 | 0 | | -1 | | | 73 | 0 | 0 |
| PERSPEX ACRYLAAT PLAAT XT GLASHELDER : | 3050 | 2050 | 2 | 100.0020 | 0 | 16-3 dr webshop | 0 | 15-3-2021 15:45:49 | 004 | 0 | 1 | 0 |
| PERSPEX ACRYLAAT PLAAT XT GLASHELDER : | 3050 | 2050 | 2 | 100.0020 | 0 | 16-3 dr webshop | 0 | 15-3-2021 15:45:49 | 005 | 0 | 1 | 0 |
| PERSPEX ACRYLAAT PLAAT XT GLASHELDER : | 3050 | 2050 | 2 | 100.0020 | 0 | 16-3 dr webshop | 0 | 15-3-2021 15:45:49 | 006 | 0 | 1 | 0 |
| PERSPEX ACRYLAAT PLAAT XT GLASHELDER : | 2050 | 1440 | 3 | | 1 | | -1 | | | 1 | 0 | 0 |
| PERSPEX ACRYLAAT PLAAT XT GLASHELDER : | 3050 | 1310 | 3 | | 1 | | -1 | | | 1 | 0 | 0 |
| PERSPEX ACRYLAAT PLAAT XT GLASHELDER : | 3050 | 2050 | 3 | 100.0030 | 0 | | -1 | | | 39 | 0 | 0 |

Views are shared through ODBC. DSS uses a Firebird database by default. Installation of the Firebird ODBC driver and the required connection settings are listed in the appendix. [➔ *Configuring database link*]

For more information about the Database link, please contact DSS Automation.

# APPENDIX

These appendices contain examples and details of imports and exports.

## Batch-xml example

**Extensive example for labeling plus nesting.**

Marked nodes are described below.

```
<?xml version="1.0" encoding="UTF-8"?>
<DSSBatch>
  <BatchName>8013009D</BatchName>
  <BatchDescription>2161151 Tramremise Nieuwegein</BatchDescription>
  <PlanDate>2018-11-05</PlanDate>
  <Priority>50</Priority>
  <BatchFlow>0</BatchFlow>
  <MaterialId>1234</MaterialId>
  <Material>SP18</Material>
  <Color>U960</Color>
  <HasGrain>False</HasGrain>
  <Filter1 />
  <Filter2 />
  <Filter3 />
  <Filter4 />
  <Filter5 />
  <Filter6 />
  <Filter7 />
  <Filter8 />
  <Filter9 />
  <MaxStackHeight>42</MaxStackHeight>
  <MaxStackBoards>1</MaxStackBoards>
  <EqualStackHeight>True</EqualStackHeight>
  <AssignBarcodes>True</AssignBarcodes>
  <AutoExpand>No</AutoExpand>
  <Parameters>
    <![CDATA[POSTPROCESSOR=COPYLABELS
POSTPROCESSOR=COPYCTS
POSTPROCESSOR=SHEETONLABELSTATION
]]>
  </Parameters>
  <PlanCount>4</PlanCount>
  <BoardCount>5</BoardCount>
  <PartCount>39</PartCount>
  <CutCount>71</CutCount>
  <Plans>
    <Plan>
      <PlanName>001</PlanName>
      <PlanOrder>1</PlanOrder>
      <Material>MDB2-19.6-U650 09</Material>
      <Color />
      <MaterialId />
      <ArticleCode />
      <Barcode />
```

```
      <XDimSize>3050</XDimSize>
      <YDimSize>1300</YDimSize>
      <ZDimSize>19.6</YDimSize>
      <Grain>1</Grain>
      <Filter1 />
      <Filter2 />
      <Filter3 />
      <Filter4 />
      <Filter5 />
      <Filter6 />
      <Filter7 />
      <Filter8 />
      <Filter9 />
      <Info2 />
      <Rotation>0</Rotation>
      <SchemaBarcode>8013009D_001</SchemaBarcode>
      <Quantity>1</Quantity>
      <SheetBarcodes>
        <SheetBarcode>8013009D_001_001</SheetBarcode>
      </SheetBarcodes>
      <CNCFilenames>
        <CNCFilename>Z:\CNC\C8013009D-1.ISO</CNCFilename>
      </CNCFilenames>
      <LabelFilenames>
        <LabelFilename>Z:\LABEL\8013009D_001_001.xml</LabelFilename>
      </LabelFilenames>
      <Parameters />
    </Plan>
    <Plan>
      <PlanName>002</PlanName>
      <PlanOrder>1</PlanOrder>
      <XDimSize>3050</XDimSize>
      <YDimSize>1300</YDimSize>
      <Grain>1</Grain>
      <SchemaBarcode>8013009D_002</SchemaBarcode>
      <Quantity>1</Quantity>
      <SheetBarcodes>
        <SheetBarcode>8013009D_002_001</SheetBarcode>
      </SheetBarcodes>
      <CNCFilenames>
        <CNCFilename>Z:\CNC\C8013009D-2.ISO</CNCFilename>
      </CNCFilenames>
      <LabelFilenames>
        <LabelFilename>Z:\LABEL\8013009D_002_001.xml</LabelFilename>
      </LabelFilenames>
    </Plan>
    <Plan>
      <PlanName>003</PlanName>
      <XDimSize>3050</XDimSize>
      <YDimSize>1300</YDimSize>
      <Grain>0</Grain>
      <SchemaBarcode>8013009D_003</SchemaBarcode>
      <Quantity>1</Quantity>
```

```
      <SheetBarcodes>
        <SheetBarcode>8013009D_003_001</SheetBarcode>
      </SheetBarcodes>
      <CNCFilenames>
        <CNCFilename>Z:\CNC\C8013009D-3.ISO</CNCFilename>
      </CNCFilenames>
      <LabelFilenames>
        <LabelFilename>Z:\LABEL\8013009D_003_001.xml</LabelFilename>
      </LabelFilenames>
    </Plan>
    <Plan>
      <PlanName>004</PlanName>
      <XDimSize>2800</XDimSize>
      <YDimSize>2070</YDimSize>
      <Material>SP18</Material>
      <Color>U960</Color>
      <Grain>0</Grain>
      <SchemaBarcode>8013009D_004</SchemaBarcode>
      <Quantity>2</Quantity>
      <SheetBarcodes>
        <SheetBarcode>8013009D_004_001</SheetBarcode>
        <SheetBarcode>8013009D_004_002</SheetBarcode>
      </SheetBarcodes>
      <CNCFilenames>
        <CNCFilename>Z:\CNC\C8013009D-4.ISO</CNCFilename>
        <CNCFilename>Z:\CNC\C8013009D-4.ISO</CNCFilename>
      </CNCFilenames>
      <LabelFilenames>
        <LabelFilename>Z:\LABEL\8013009D_004_001.xml</LabelFilename>
        <LabelFilename>Z:\LABEL\8013009D_004_002.xml</LabelFilename>
      </LabelFilenames>
    </Plan>
  </Plans>
  <Routes>
    <Route>
      <Order>1</Order>
      <Name>Stock-to-Label</Name>
      <FromStackposition />
      <ToStackposition>LABEL</ToStackposition>
      <Rotation>0</Rotation>
      <Parameters />
    </Route>
    <Route>
      <Order>2</Order>
      <ToStackposition>CNC</ToStackposition>
    </Route>
  </Routes>
</DSSBatch>
```

DSSBatch / root node:

- BatchName : Name/number/reference for this production order (cutting, milling, …).
- Name : <see BatchName> [deprecated].

- BatchDescription : Description of the production order.
- Description1 : <see BatchDescription> [deprecated].
- Description : <see BatchDescription> [deprecated].
- PlanDate : Date the production is scheduled to be done (yyyy-mm-dd). [optional; default=file date]
- Date : equal to PlanDate [deprecated].
- Priority : Priority of the batch (1=low, 99=high). [optional; default=70]
- BatchFlow : Parameter to describe the behavior of the batch:
    - 0 = ByPiece. Each stack of sheets is handled route step by route step, until all steps are done. Then the next stack is processed. Etc. [default].
    - 1 = ByBundle : Each route step is handled for all sheets of the batch, then the next route step, until all steps are done.
    - 2 = ByBatch : Same as ByBundle, except that transport works are created *by batch* instead of *by job*.
- MaterialId : ID of the material. If present in the DSSBatch-node, this material will be used as default for the Plan-nodes. If this ID is empty, it is derived from the Material- and Color-nodes [optional].
- Material : Material name. If present in the DSSBatch-node, this material will be used as default for the Plan-nodes [optional].
- Color: Material color. If present in the DSSBatch-node, this color will be used as default for the Plan-nodes [optional].
- HasGrain : Indicates the material has a veneer or pattern direction. (Boolean; True, Yes, No, False) [optional; default=False].
- Filter1 : Extra label for this batch, e.g. the name of the customer. This data is used to match the sheets for this batch.
- Filter2 : <see Filter1>
- Filter3 : <see Filter1>
- Filter4 : <see Filter1>
- Filter5 : <see Filter1>
- Filter6 : <see Filter1>
- Filter7 : <see Filter1>
- Filter8 : <see Filter1>
- Filter9 : <see Filter1>
- Info1 : equal to Filter1 [deprecated].
- Info2 : equal to Filter2 [deprecated].
- Info3 : equal to Filter3 [deprecated].
- MaxStackHeight : Maximum loading height on the saw/nesting/… machines. (mm) [optional; default=0].
- MaxStackBoards : Maximum loading height on the saw/nesting/… machines. (pieces) [optional; default=0].
- EqualStackHeight : Indicates how to split stacked sheets when moving more than one. (Boolean; True, Yes, No, False) [optional; default=False]
    - True = keep stacks at the same height, when possible
        - e.g.: 7 sheets, maxstackboards=5 → 4+3

    - False = stack as high as possible
        - e.g.: 7 sheets, maxstackboards=5 → 5+2
- AssignBarcodes : Indicates whether sheets should be assigned a barcode when processing the batch. (Boolean; True, Yes, No, False) [optional; default=True].
- AutoExpand : Indicates if the batch has to be started automatically. [optional]
    - No : Batch has to be started manually. [default]
    - Yes : Batch is started automatically as soon as there are no batches in status Error and no other batches for the same routing are running or paused.
    - Semi : AutoExpand is set to “Yes” when the user starts the batch.
    - Soon : Batch is started automatically as soon as there are no batches with AutoExpand=Soon in status Error and nu other batches for the same routing running.
    - Direct : Batch is started immediately after import.
- Parameters: Specific settings for pre- or postprocessing. Contact DSS Automation for details.
- PlanCount : Number of different schemas in this order [optional].
- BoardCount : Total number of sheets used in this order [optional].
- PartCount : Number of different parts in this order [optional].
- CutCount : Total number of cut parts in this order [optional].

*Note*: MaxStackHeight / MaxStackBoards. A stack always consists of 1 or more sheets. These settings describe at what height (in mm or in pieces) a stack of sheets should be split.

Plan nodes:

- PlanName : Name / number of this schema / layout. If empty, it is automatically filled with “001”, “002”, “003”, etc. based on the order the Plan-nodes appear in the xml-file [optional].
- PlanOrder : The order in which the batch is to be processed. If empty, the order is based on the alphabetic order of the PlanNames [optional].
- ArticleCode : Article code of the article to be used. If present, the Material, Length, Width and Grain are derived from this Articlecode [optional].
- Barcode : Barcode of the article to be used. If present, the Material, Length, Width and Grain are derived from this barcode [optional]. *Note*: this is a reference to the article (MaterialSize table), not to the sheet (Sheet table).
- MaterialId : ID of the material. If not present, the default from the DSSBatch-node will be used. If this ID is empty, it is derived from the Material- and Color-nodes [optional].
- Material : Material name. If not present, the default from the DSSBatch-node is used [optional].
- Color: Material color. If not present, the default from the DSSBatch-node will be used [optional].
- XDimSize : Length of the sheet (mm, maximum 1 decimal; dot as decimal separator). [optional is ArticleCode or Barcode is used]
- YDimSize : Width of the sheet (mm, maximum 1 decimal; dot as decimal separator). [optional is ArticleCode or Barcode is used]
- ZDimSize : Thickness of the sheet. Default is the thickness of the associated material (mm, maximum 1 decimal; dot as decimal separator) [optional].
- Grain : Grain direction of the sheet: 0=None; 1=Long (parallel to the length); 2=Short (parallel to the width). [optional is ArticleCode or Barcode is used]; default=0]

- Filter1 : Extra attribute for this plan, e.g. the name of the customer. This data is used to match the sheets for this plan and do stock reservations [optional].
- Filter2 : <see Filter1>
- Filter3 : <see Filter1>
- Filter4 : <see Filter1>
- Filter5 : <see Filter1>
- Filter6 : <see Filter1>
- Filter7 : <see Filter1>
- Filter8 : <see Filter1>
- Filter9 : <see Filter1>
- Info1 : equal to Filter1 [deprecated].
- Info2 : equal to Filter2. *Note*: "Info2" is also used for composing the *display name* of the material used in this plan. It has in this context, however, no influence on the linked material.
- Info3 : equal to Filter3 [deprecated].
- Rotation : Rotation angle of the sheet for all movements. [optional; default=0].
- SchemaBarcode : Code of the schema for this sheet [optional].
- Quantity : Number of sheets for this plan.
- SheetBarcode : Barcode (= unique reference) to the sheet used for this plan. This barcode can be generated automatically by DSS during import (based on labelfilename, cncfilename or batchname+planname+index), depending on the setting of the AssignBarcodes-node [optional].
- CNCFilenames : List of CNCFilename [only required for nesting].
- CNCFilename : Name of the cnc-file [only required for nesting].
- LabelFilenames : List of LabelFilename [only required for labeling].
- LabelFilename : Name of the label file. If a LabelFilename is present, DSSClient will look for a bitmap (Portable Network Graphics (*.png) or Windows Bitmap (*.bmp)) with the same name and in the same folder as the labelfile to serve as a preview for the saw/cnc-pattern. [only required for labeling].
- Parameters: specific settings for pre- or postprocessing. Contact DSS Automation for details.

Note: To determine the material of the sheet, either *MaterialId* or *Material* (with optional *Color*) or *ArticleCode* or *Barcode* is required.

Note: *XDimSize* should always be larger than or equal to *YDimSize*. If not, DSS will swap both values, and change the value for *Grain* accordingly, automatically.

Route nodes:

- Order : Route-index. If empty it is based on the order the Route-nodes appear in the xml-file [optional].
- Name : Name of the step [optional].
- FromStackposition : The ID, Alias or Name of the position the sheet is to be taken from in this step. Mandatory for batches used for stock-in-operations, optional in all other cases.

- ToStackPosition : The ID, Alias or Name of the position / machine the sheets have to be moved to in this step. Optional for batches used for stock-in-operations, mandatory in all other cases.
- Rotation : Additional rotation angle of the sheet for this step. [optional; default=0].
- Parameters: specific settings for pre- or postprocessing. Contact DSS Automation for details.

(Nodes / attributes that are not explicitly described here are to be considered static)

**Simple example for sawing:**

This file loads two small stacks of sheets onto the saw. No labeling is involved. After loading a stack, the saw is sent instructions to load and start the corresponding saw schema.

```
<?xml version="1.0" encoding="utf-8"?>
<DSSBatch>
  <BatchName>2100012C</BatchName>
  <BatchDescription>JOY INTERIEUR 21003790</BatchDescription>
  <PlanDate>2021-03-19</PlanDate>
  <MaxStackHeight>60</MaxStackHeight>
  <EqualStackHeight>True</EqualStackHeight>
  <Parameters>
    <![CDATA[POSTPROCESSOR=CopyInfeed
]]>
</Parameters>
  <PlanCount>2</PlanCount>
  <BoardCount>3</BoardCount>
  <PartCount>6</PartCount>
  <CutCount>8</CutCount>
  <Plans>
    <Plan>
      <PlanName>001</PlanName>
      <Material>SP 16.5 + 4167WH/4167WH</Material>
      <XDimSize>3050</XDimSize>
      <YDimSize>1320</YDimSize>
      <Grain>1</Grain>
      <SchemaBarcode>2100012C_001</SchemaBarcode>
      <Quantity>1</Quantity>
      <Rotation>0</Rotation>
    </Plan>
    <Plan>
      <PlanName>002</PlanName>
      <Material>SP 18 W910</Material>
      <XDimSize>2800</XDimSize>
      <YDimSize>2070</YDimSize>
      <Grain>0</Grain>
      <SchemaBarcode>2100012C_002</SchemaBarcode>
      <Quantity>2</Quantity>
      <Rotation>0</Rotation>
    </Plan>
  </Plans>
  <Routes>
    <Route>
      <ToStackPosition>SAW</ToStackPosition>
    </Route>
  </Routes>
</DSSBatch>
```

**Simple example for order picking:**

A set of four sheets is stored in the IN-UIT position. When done, a packing slip is printed with a list of the sheets in the stack.

```
<?xml version="1.0" encoding="utf-8"?>
<DSSBatch>
  <BatchName>2400042</BatchName>
  <BatchDescription>HH JANSSEN 2400042</BatchDescription>
  <PlanDate>2024-05-19</PlanDate>
  <Parameters>
    <![CDATA[POSTPROCESSOR=PrintPackingSlip
]]>
</Parameters>
  <PlanCount>3</PlanCount>
  <BoardCount>4</BoardCount>
  <Plans>
    <Plan>
      <PlanName>001</PlanName>
      <Material>DEKPLAAT</Material>
      <XDimSize>3050</XDimSize>
      <YDimSize>1200</YDimSize>
      <Grain>0</Grain>
      <Quantity>1</Quantity>
    </Plan>
    <Plan>
      <PlanName>002</PlanName>
      <Material>SP 12.0 9010</Material>
      <XDimSize>3050</XDimSize>
      <YDimSize>1200</YDimSize>
      <Grain>1</Grain>
      <Quantity>1</Quantity>
    </Plan>
    <Plan>
      <PlanName>003</PlanName>
      <Material>MDF 18</Material>
      <XDimSize>3050</XDimSize>
      <YDimSize>1300</YDimSize>
      <Grain>0</Grain>
      <Quantity>2</Quantity>
    </Plan>
  </Plans>
  <Routes>
    <Route>
      <ToStackPosition>IN-UIT</ToStackPosition>
    </Route>
  </Routes>
</DSSBatch>
```

---

## Batch-lst example

A batch list is a tab-separated text file with the “LST” extension. Each line contains all the details required for handling one or more identical sheets. The file always starts with a header line naming the data columns. For example (marked column names are described below):

```
BATCH.BATCHNAME→BATCH.BATCHDESCRIPTION→MATERIAL→LENGTH→WIDTH→CNCFILENAME
→LABELFILENAME→SHEETBARCODE→QUANTITY→ROUTING↵
20201028A→Nesting_A→SH_RX_PC_PC_CLEAR_6.0_NO_NA→3050→2050→C:\20201028A01
.cix→C:\20201028A01_lbls.xml→20201028A_01→1→LABEL,SAW↵
20201028A→Nesting_A→SH_RX_PC_PC_CLEAR_8.0_NO_NA→3050→2050→C:\20201028A02
.cix→C:\20201028A02_lbls.xml→20201028A_02 →1→LABEL,SAW↵
```

*Note*: this example consists of one header line and two data lines. The "→" marks the tab-symbol (which would otherwise be invisible); the "↵" denotes the end of a line, for the same reason. These symbols are used here for clarity only.

BATCH columns:

- BATCH.BATCHNAME : Name/number/reference for this production order (sawing, cutting, milling, …).
- BATCH.BATCHDESCRIPTION : Description of the production order.
- BATCH.PLANDATE : Date the production is scheduled to be done (yyyy-mm-dd). [default=file date]
- BATCH.MATERIAL : Material name. [optional].
- BATCH.COLOR : Material color. [optional]
- BATCH.MATERIALID : ID of the material. If this ID is empty, it is derived from the BATCH.MATERIAL and BATCH.COLOR columns. [optional].
- BATCH.HASGRAIN : Indicates the material has a veneer or pattern direction. (Boolean; True, Yes, 1, No, False, 0) [optional; default=False].
- BATCH.MAXSTACKHEIGHT : Maximum loading height on the saw/nesting/… machines. (mm) [optional; default=0].
- BATCH.MAXSTACKBOARDS : Maximum loading height on the saw/nesting/… machines. (pieces) [optional; default=0].
- BATCH.EQUALSTACKHEIGHT : Indicates how to split stacked sheets when moving more than one. (Boolean; True, Yes, 1, No, False, 0) [optional; default=False]
    - True = keep stacks at the same height, when possible
        - e.g.: 7 sheets, maxstackboards=5 → 4+3
    - False = stack as high as possible
        - e.g.: 7 sheets, maxstackboards=5 → 5+2
- BATCH.PLANCOUNT : Number of different schemas in this order.
- BATCH.BOARDCOUNT : Total number of sheets used in this order.
- BATCH.PARTCOUNT : Number of different parts in this order [optional].
- BATCH.CUTCOUNT : Total number of cut parts in this order [optional].
- BATCH.AUTOEXPAND : Indicates if the batch has to be started automatically. [optional]
    - No : Batch has to be started manually. [default]
    - Yes : Batch is started automatically as soon as there are no batches in status Error and no other batches for the same routing are running or paused.
    - Semi : AutoExpand is set to "Yes" when the user starts the batch.
    - Soon : Batch is started automatically as soon as there are no batches with AutoExpand=Soon in status Error and nu other batches for the same routing running.

    - Direct : Batch is started immediately after import.
- BATCH.BATCHFLOW : Parameter to describe the behavior of the batch:
    - 0 = ByPiece. Each stack of sheets is handled route step by route step, until all steps are done. Then the next stack is processed. Etc. [default].
    - 1 = ByBundle : Each route step is handled for all sheets of the batch, then the next route step, until all steps are done.
    - 2 = ByBatch : Same as ByBundle, except that transport works are created *by batch* instead of *by job*.
- BATCH.PRIORITY : Priority of the batch (1=low, 99=high). [optional; default=70]
- BATCH.PARAMETERS : Specific settings for pre- or postprocessing. Contact DSS Automation for details [optional].

*Note*: BATCH.MAXSTACKHEIGHT / BATCH.MAXSTACKBOARDS. A stack always consists of 1 or more sheets. These settings describe at what height (in mm or in pieces) a stack of sheets should be split.

PLAN columns:

- PLAN.MATERIAL : equal to BATCH.MATERIAL.
- PLAN.COLOR : equal to BATCH.COLOR.
- PLAN.MATERIALID : equal to BATCH.MATERIALID.
- PLAN.XDIMSIZE : Length of the sheet (mm, maximum 1 decimal; dot as decimal separator).
- PLAN.LENGTH : equal to PLAN.XDIMSIZE.
- PLAN.YDIMSIZE : Width of the sheet (mm, maximum 1 decimal; dot as decimal separator).
- PLAN.WIDTH : equal to PLAN.YDIMSIZE.
- PLAN.ROTATION : Rotation angle of the sheet for all movements. [optional; default=0].
- PLAN.GRAIN : Grain direction of the sheet: 0=None; 1=Long (parallel to the length); 2=Short (parallel to the width). [optional; default=0]
- PLAN.SCHEMABARCODE : Code of the schema for this sheet [optional].
- PLAN.QUANTITY : Number of sheets for this plan. [optional; default=1]
- PLAN.SHEETBARCODE : Barcode (= unique reference to) to the sheet used for this plan. This barcode can be generated automatically by DSS during import (based on labelfilename, cncfilename or batchname+planname+index), depending on the setting of the AssignBarcodes-node [optional].
- PLAN.SHEETBARCODES : equal to PLAN.SHEETBARCODE.
- PLAN.LABELFILENAME : Semi-colon separated list of the name(s) of the label file(s) [optional].
- PLAN.LABELFILENAMES : equal to PLAN.LABELFILENAME.
- PLAN.CNCFILENAME : Semi-colon separated list of the name(s) of the cnc-file(s) [optional].
- PLAN.CNCFILENAMES : equal to PLAN.CNCFILENAME.
- PLAN.PARAMETERS : Specific settings for pre- or postprocessing. Contact DSS Automation for details [optional].

ROUTE columns:

- ROUTING : Comma-separated list of to-stackpositions. Each item in the list is one route step. Special cases:
    - The to-stackposition is a semi-colon-separated list: the operator can choose what stack position to use when starting the batch.
    - The to-stackposition starts with a question mark: the operator has to choose what stack position to use when starting the batch.
    - The to-stackposition is a forward-slash-separated list: when running the batch, sheets will be moved to the to-stackpositions in the list alternating between them.
- ROUTE.PARAMETERS : Specific settings for pre- or postprocessing. Contact DSS Automation for details [optional].

If columns are to contain the same values for all files that are imported (e.g. BATCH.MAXSTACKHEIGHT=80 or ROUTING=SAW), then the column can be omitted and the default value added to the import settings.

Column names can be preceded with the “BATCH.” , “PLAN.” or “ROUTE.” identifier, but this is not mandatory.

---

## CTS example

As of writing, DSS natively support imports of the following CTS file formats:

- Biesse BWKL-files (including label information)
- Biesse WLST-files
- Biesse CIX-files
- Biesse Selco / Optiplanning XML-files
- Holzma Pattern Exchange / PTX-files
- Holzma Aktplan-files
- Holz-her NCR-files
- Mayer MAY-files
- Mazak NC-files
- Trumpf LST-files

Details are available through the corresponding companies.

---

## Picklist-xml example

Marked nodes are described below.

```
<?xml version="1.0" encoding="utf-8"?>
<DSSPicklist>
  <PicklistName>202010.321</PicklistName>
  <PicklistDescription>3x Belgian Blue 30 (volledig)</PicklistDescription>
  <PlanDate>2020-09-18</PlanDate>
  <Stackposition>1002</Stackposition>
  <PickMode>0</PickMode>
  <Priority>10</Priority>
```

```
<Parameters>POSTPROCESSOR=Picklistdone</Parameters>
<Lines>
  <Line>
    <LineNo>10</LineNo>
    <MaterialId>5617</MaterialId>
    <Material>BU007/BELGIAN BLUE BUV/30</Material>
    <Color>WHITE</Color>
    <XDimSize>3000</XDimSize>
    <YDimSize>1400</YDimSize>
    <Grain>0</Grain>
    <ArticleCode>BB3821</ArticleCode>
    <Barcode />
    <Filter1>K</Filter1>
    <Filter2>GEZOET, MAT</Filter2>
    <Filter3>II07</Filter3>
    <Filter4 />
    <Filter5 />
    <Filter6 />
    <Filter7 />
    <Filter8 />
    <Filter9 />
    <SheetBarcode>80II07C076G</SheetBarcode>
    <Quantity>1</Quantity>
  </Line>
  <Line>
    <LineNo>20</LineNo>
    <MaterialId>5617</MaterialId>
    <Material>BU007/BELGIAN BLUE BUV/30</Material>
    <Color></Color>
    <XDimSize>3000</XDimSize>
    <YDimSize>1400</YDimSize>
    <Grain>0</Grain>
    <ArticleCode></ArticleCode>
    <Filter1>K</Filter1>
    <Filter2>GEZOET, MAT</Filter2>
    <Filter3>JD26</Filter3>
    <SheetBarcode>80JD26C321G</SheetBarcode>
    <Quantity>1</Quantity>
  </Line>
  <Line>
    <LineNo>30</LineNo>
    <MaterialId>5617</MaterialId>
    <Material>BU007/BELGIAN BLUE BUV/30</Material>
    <Color></Color>
    <XDimSize>3000</XDimSize>
    <YDimSize>1400</YDimSize>
    <Grain>0</Grain>
    <ArticleCode></ArticleCode>
    <Filter1>K</Filter1>
    <Filter2>BLINKEND</Filter2>
    <Filter3>JA17</Filter3>
    <SheetBarcode>80JA17C141F</SheetBarcode>
    <Quantity>1</Quantity>
```

```
    </Line>
  </Lines>
</DSSPicklist>
```

DSSPicklist / root node:

- PicklistName : Name/number/reference for this pick list.
- PicklistDescription : Description of the pick list
- PlanDate : Date the pick list is scheduled to be done (yyyy-mm-dd). [optional; default=file date].
- Stackposition : Name or alias of the stackposition the sheets will be stored.
- PickMode : Order in which to pick the sheets. 0=Ascending;1=Descending; 2=ByAvailability. [optional; default=0].
- Priority : Priority of the batch (1=low, 99=high). [optional; default=50].
- Parameters : Specific settings for pre- or postprocessing. Contact DSS Automation for details [optional].

Line node:

- LineNo : The order in which the pick list is to be processed. If empty, the order is based on the order the Line-nodes appear in the xml-file [optional].
- SheetBarcode : Semi-colon separated list of barcodes (= unique reference) to sheets. If present, ArticleCode, Barcode, Material, Length, Width and Grain are derived from this SheetBarcode. Use the SheetBarcode to pick a (set of) specific sheet(s) from the warehouse, not just any sheet that matches material, dimensions and grain [optional].
- SheetBarcodes : <see SheetBarcode>
- ArticleCode : Article code of the article to be used. If present, the Material, Length, Width and Grain are derived from this Articlecode [optional].
- Barcode : Barcode of the article to be used. If present, the Material, Length, Width and Grain are derived from this barcode [optional]. *Note*: this is a reference to the article (MaterialSize table), not to the sheet (Sheet table).
- MaterialId : ID of the material. If not present, it is derived from the Material- and Color-nodes [optional].
- Material : Material name. [optional].
- Color: Material color. [optional].
- XDimSize : Length of the sheet (mm, maximum 1 decimal; dot as decimal separator). [optional is ArticleCode or Barcode is used]
- YDimSize : Width of the sheet (mm, maximum 1 decimal; dot as decimal separator). [optional is ArticleCode or Barcode is used]
- Grain : Grain direction of the sheet: 0=None; 1=Long (parallel to the length); 2=Short (parallel to the width). [optional is ArticleCode or Barcode is used]; default=0].
- Filter1 : Extra attribute for this sheet, e.g. the name of the customer [optional].
- Filter2 : <see Filter1>
- Filter3 : <see Filter1>
- Filter4 : <see Filter1>
- Filter5 : <see Filter1>

- Filter6 : <see Filter1>
- Filter7 : <see Filter1>
- Filter8 : <see Filter1>
- Filter9 : <see Filter1>
- Quantity : Number of sheets.

Note: To determine the material of the sheet, *MaterialId* or *Material* (with optional *Color*) or *ArticleCode* or *Barcode* or *SheetBarode* is required.

Note: *XDimSize* should always be larger than or equal to *YDimSize*. If not, DSS will swap both values, and change the value for *Grain* accordingly, automatically.

**Example of a simple and strict picklist-xml (using only SheetBarcode):**

```
<?xml version="1.0" encoding="utf-8"?>
<DSSPicklist>
  <Name>202010.321</Name>
  <Lines>
    <Line>
      <SheetBarcode>80II07C076G;80JD26C321G</SheetBarcode>
      <Quantity>2</Quantity>
    </Line>
  <Lines>
</DSSPickList>
```

In this example, only the two sheets with these exact barcodes are collected.

**Example of a simple but more flexible picklist-xml (using material and sheet dimensions):**

```
<?xml version="1.0" encoding="utf-8"?>
<DSSPicklist>
  <Name>202010.321</Name>
  <Lines>
    <Line>
      <Material>BU007/BELGIAN BLUE BUV/30</Material>
      <XDimSize>3000</XDimSize>
      <YDimSize>1400</YDimSize>
      <Grain>0</Grain>
      <Quantity>2</Quantity>
    </Line>
  </Lines>
</DSSPickList
```

In this example, two sheets with the specified material and dimensions are collected. DSS will find the proper sheets that cost the least effort to obtain, to prevent unnecessary digging.

---

## Picking-xml (batch) example

For a description of a batch-xml for picking, <see Batch-xml example>. That paragraph ends with a "*Simple example for order picking*".

The Routes-node describes where the sheets have to be taken to, optionally in different steps along a route. For a picking order, this is just one step. The destination might be fixed, as with a regular batch-xml, or a list of options to choose from:

- ToStackposition : The output station(s) where all the sheets of this picking order have to be dropped off. The value contains a single station name or a list of station names, separated by a semi-colon (";"). When a list of stations is present, DSS will determine which station to use based on availability. If a list is present *and* it is preceded by a question mark ("?"), the operator has to select the output station when starting the picking order.
  e.g. `<ToStackPosition>INUIT3</ToStackPosition>`
  e.g. `<ToStackPosition>INUIT2;INUIT3</ToStackPosition>`)
  e.g. `<ToStackPosition>?PALLET1;PALLET2;PALLET3</ToStackPosition>`

---

## Recipe-xml example

t.b.d.

---

## Label-csv example

The csv contains coordinates of the label relative to the sheet, and data – or a reference to a picture – to print on the label. The data is merged with a label template to produce actual labels.

| LENGTH | WIDTH | ID | XPOS | YPOS | ROTATION | IMAGE |
|---|---|---|---|---|---|---|
| 2820 | 2070 | 128931 | 1145 | 299 | 0 | Y:\DSS\Label\12893_1.bmp |
| 2820 | 2070 | 128933 | 1151 | 891 | 90 | Y:\DSS\Label\12893_3.bmp |
| 2820 | 2070 | 128932 | 1157 | 1483 | 0 | Y:\DSS\Label\12893_2.bmp |
| … | … | … | … | … | … | … |

A label-csv is a tab, comma or semi-colon-separated file consisting of a header line and for each label one data line.

Columns:

- LENGTH : Length of the sheet (mm, no decimals).
- WIDTH : Length of the sheet (mm, no decimals).
- ID : Unique reference to the label. [optional]
- XPOS : X-coordinate of the label. The x-axis is always parallel to the length of the sheet. (mm, no decimals).
- YPOS : Y-coordinate of the label. The y-axis is always parallel to the width of the sheet. (mm, no decimals).
- ROTATION : Rotation angle of the label. 0, 1 or a multiple of 90. [optional; default=0]

The data to print on the labels is provided as extra columns. In the example above, this is column "IMAGE".

The label template itself is an xml-file containing a mix of print objects [➔ *Label-xml example*] and variables referring to columns in the csv-file. The variables are defined as '$' + column name + '$' , e.g. '$LENGTH$'. A label template describes only one <PrintLabel> node. This node is repeated for each data line in the csv-file during the merge.

**Label template example with only one image object:**

```
<?xml version="1.0" encoding="UTF-8"?>
<DSSPrintJob Length="$LENGTH$" Width="$WIDTH$">
  <PrintLabel ID="$INDEX$" X="$XPOS$" Y="$YPOS$" Rotation="$ROTATE$">
    <Image X="20" Y="20" X2="80" Y2="80">$IMAGE$</Image>
  </PrintLabel>
</DSSPrintJob>
```

Label-csv with extra data columns:

| LENGTH | WIDTH | ID | XPOS | YPOS | ROTATION | ORDERNO | PW | PH | BARCODE1 |
|---|---|---|---|---|---|---|---|---|---|
| 2820 | 2070 | 931 | 1145 | 299 | 0 | 3629310 | 800 | 600 | 3629310-931 |
| 2820 | 2070 | 933 | 1151 | 891 | 90 | 3629310 | 788 | 600 | 3629310-933 |
| 2820 | 2070 | 932 | 1157 | 1483 | 0 | 3629310 | 788 | 350.2 | 3629310-932 |
| … | … | … | … | … | … | … | … | … | … |

**Label template example with multiple print objects:**

```
<?xml version="1.0" encoding="UTF-8"?>
<DSSPrintJob Length="$LENGTH$" Width="$WIDTH$"">
  <PrintLabel ID="$ID$" X="$XPOS$" Y="$YPOS$" ROTATION="$ROTATION$">
    <Line X="70" Y="15" X2="70" Y2="90"/>
    <Text X="10" Y="15">Project:</Text>
    <Text X="25" Y="15" Style="BOLD">$ORDERNO$</Text>
    <Text X="10" Y="25">Size:</Text>
    <Text X="25" Y="25">$PW$</Text>
    <Text X="45" Y="25">x</Text>
    <Text X="65" Y="25">$PH$</Text>
    <Line X="10" Y="58" X2="70" Y2="58"/>
    <Image X="10" Y="75" X2="50" Y2="90" >C:\DSS\DUIVESTEIN.png</Image>
    <Barcode X="75" Y="10" Width="70" Height="15" Size="2"
Type="BARCODE128A" Rotation="270">$BARCODE1$</Barcode>
    <Arrow X="75" Y="75" X2="90" Y2="90" Direction="7" Weight="2"
FillStyle="7" />
  </PrintLabel>
</DSSPrintJob>
```

---

## Label-xml example

A label-xml file describes the content and coordinates of all labels on a sheet. Marked nodes are described below.

```
<?xml version="1.0" encoding="utf-8"?>
<DSSPrintJob Length="2800" Width="2100">
  <PrintLabel ID="5436280" X="500" Y="1000">
    <Image X="6" Y="6" X2="94" Y2="33">C:\DSS\DUIVESTEIN-LOGO.bmp</Image>
    <Line X="5" Y="45" X2="95" Y2="45" />
    <Text X="7" Y="47" FontSize="10" Font="Arial">Materiaal :</Text>
    <Text X="30" Y="47" FontSize="10" Font="Arial">MDF_D18</Text>
    <Text X="7" Y="52" FontSize="10" Font="Arial">Afmetingen :</Text>
    <Text X="30" Y="52" FontSize="10" Font="Arial">280 x 421 x 18</Text>
    <Text X="7" Y="57" FontSize="10" Font="Arial">Project :</Text>
    <Text X="30" Y="57" FontSize="10" Font="Arial">BEURS2017 / 001.A</Text>
    <Text X="7" Y="62" FontSize="10" Font="Arial">Klant :</Text>
    <Text X="30" Y="62" FontSize="10" Font="Arial">Kerssemakers</Text>
    <Line X="5" Y="69" X2="95" Y2="69" />
    <Text X="7" Y="71" FontSize="10" Font="Arial">Datum / Tijd :</Text>
    <Text X="30" Y="71" FontSize="10" Font="Arial">$DATE$ / $TIME$</Text>
    <Barcode X="5" Y="80" Width="90" Rotation="0"
    Type="BARCODE128A" Size="7" Height="15">13579-houtproplus-
24680</Barcode>
  </PrintLabel>
  <PrintLabel ID="5436281" X="650" Y="300">
    <Image X="6" Y="6" X2="94" Y2="33">C:\DSS\DUIVESTEIN-LOGO.bmp</Image>
    <Line X="5" Y="45" X2="95" Y2="45" />
    <Text X="7" Y="47" FontSize="10" Font="Arial">Materiaal :</Text>
    <Text X="30" Y="47" FontSize="10" Font="Arial">MDF_D18</Text>
    <Text X="7" Y="52" FontSize="10" Font="Arial">Afmetingen :</Text>
    <Text X="30" Y="52" FontSize="10" Font="Arial">680 x 800 x 18</Text>
    <Text X="7" Y="57" FontSize="10" Font="Arial">Project :</Text>
    <Text X="30" Y="57" FontSize="10" Font="Arial">BEURS2017 / 001.A</Text>
    <Text X="7" Y="62" FontSize="10" Font="Arial">Klant :</Text>
    <Text X="30" Y="62" FontSize="10" Font="Arial">Kerssemakers</Text>
    <Line X="5" Y="69" X2="95" Y2="69" />
    <Text X="7" Y="71" FontSize="10" Font="Arial">Datum / Tijd :</Text>
    <Text X="30" Y="71" FontSize="10" Font="Arial">$DATE$ / $TIME$</Text>
    <Barcode X="5" Y="80" Width="90" Rotation="0"
    Type="BARCODE128A" Size="7" Height="15">13579-houtproplus-
53210</Barcode>
  </PrintLabel>
</DSSPrintJob>
```

DSSPrintJob / root nodes:

- Length : Length of the sheet (mm, no decimals).
- Width : Width of the sheet (mm, no decimals).
- ID : Unique reference to the label. [optional]
- X : X-coordinate of the label on the sheet (mm, no decimals).
- Y : Y-coordinate of the label on the sheet (mm, no decimals).

Each label consists of one or more “print objects” with attributes:

- Text : A test object to display alphanumeric data. Attributes:

  - X : X-coordinate (mm, dot as decimal separator).
  - Y : Y- coordinate (mm, dot as decimal separator).
  - Rotation : Rotation angle. [optional; default=0]
  - Font : Font name. [optional; default=defined by DSSPrintServer]
  - FontSize : Height of the font (points).
  - FontStyle : Semi-colon-separated list of font style keywords BOLD, ITALIC, UNDERLINE and STRIKEOUT. [optional]
  - <Value> : Text to print. The variables $DATE$ and $TIME$ are replaced with the current date and time as the label is printed.
- Line : A line object. Attributes:
  - X : X-coordinate of the starting point of the line (mm, dot as decimal separator).
  - Y : Y-coordinate of the starting point of the line (mm, dot as decimal separator).
  - X2 : X-coordinate of the endpoint of the line (mm, dot as decimal separator).
  - Y2 : Y-coordinate of the endpoint of the line (mm, dot as decimal separator).
  - Weight : Thickness of the line (points). [optional; default=1]
- Rectangle : A rectangle object. Attributes:
  - X : X-coordinate of the upper left-hand corner of the rectangle (mm, dot as decimal separator).
  - Y : Y-coordinate of the upper left-hand corner of the rectangle (mm, dot as decimal separator).
  - X2 : X-coordinate of the lower right-hand corner of the rectangle (mm, dot as decimal separator).
  - Y2 : Y-coordinate of the lower right-hand corner of the rectangle (mm, dot as decimal separator).
  - Weight : Thickness of the lines (points). [optional; default=1]
- Image : A bitmap image object. Supports Portable Network Graphics (*.png) files and Windows Bitmap (*.bmp) files. Attributes:
  - X : X-coordinate of the upper left-hand corner of the rectangle (mm, dot as decimal separator).
  - Y : Y-coordinate of the upper left-hand corner of the rectangle (mm, dot as decimal separator).
  - X2 : X-coordinate of the lower right-hand corner of the rectangle (mm, dot as decimal separator).
  - Y2 : Y-coordinate of the lower right-hand corner of the rectangle (mm, dot as decimal separator).
  - <value> : Full path name to the image, or relative path to the label-xml.
- Barcode : Linear barcode object. Attributes:
  - X : X-coordinate (mm, dot as decimal separator).
  - Y : Y- coordinate (mm, dot as decimal separator).
  - Rotation : Rotation angle. [optional; default=0]
  - Width : Width of the barcode (mm, no decimals).
  - Height : Height of the barcode (mm, no decimals).
  - Size : Barcode scale factor. [optional; default=1]

    - Type : Type of barcode. Options: BARCODE128, BARCODE128A, BARCODE128B, BARCODE128C, BARCODE39, BARCODE93, BARCODEEAN8 and BARCODEEAN13. [optional; default=BARCODE128]
    - <Value> : Barcode to print.
- QRCode : QR-code object. Attributes:
    - X : X-coordinate (mm, dot as decimal separator).
    - Y : Y- coordinate (mm, dot as decimal separator).
    - Width : Width of the barcode (mm, no decimals).
    - Height : Height of the barcode (mm, no decimals).
    - Type : Type of QR-code. Options: AUTO, NUMERIC, ALPHANUMERIC, ISO88591, UTF8NOBOM and UTF8BOM. [optional; default=AUTO]
    - <Value> : QR-code to print.
- Arrow : An arrow object. Attributes:
    - X : X-coordinate of the upper left-hand corner of the bounding box of the arrow (mm, dot as decimal separator).
    - Y : Y-coordinate of the upper left-hand corner of the bounding box of the arrow (mm, dot as decimal separator).
    - X2 : X-coordinate of the lower right-hand corner of the bounding box of the arrow (mm, dot as decimal separator).
    - Y2 : Y-coordinate of the lower right-hand corner of the bounding box of the arrow (mm, dot as decimal separator).
    - Direction : Direction the arrow points to. Options: 1=LowerLeft, 2=MidLeft, 3=TopLeft, 4=TopMid, 5=TopRight, 6=MidRight, 7=LowerRight, 8=LowerMid. [optional; default=6]
    - FillStyle : Fill pattern of the arrow. Options: 0=Solid, 1=Clear, 2=Horizontal, 3=Vertical, 4=FDiagonal, 5=BDiagonal, 6=Cross, 7=DiagCross. [optional; default=0]

To *not* print labels, an “empty” label-xml is required:

```
<?xml version="1.0" encoding="utf-8"?>
<DSSPrintJob Length="2800" Width="2100" Turned="false">
</DSSPrintJob>
```

---

## Cranejob-xml example

t.b.d.

---

## Deposit-xml example

t.b.d.

---

## Loadingdocklist-xml example

t.b.d.

---

## Restpiecelist-xml example

Example with a list of 3 sheets in total: one set with all and another single one with just the minimum number of details for a restpiece. Note that the xml-file does not specify where the rest pieces are taken from nor on what stackposition in the warehouse they are to be stored; this is defined by settings in the DSS software and input from the operator at the console. Marked nodes are described below.

```
<?xml version="1.0" encoding="utf-8"?>
<DSSRestPieceList>
  <RestPiece>
    <MaterialId>3621</MaterialId>
    <Material>SPN-OAK-18</Material>
    <Color>DARK03</Color>
    <XDimSize>2800</XDimSize>
    <YDimSize>1725.3</YDimSize>
    <ZDimSize>18</ZDimSize>
    <Grain>1</Grain>
    <Weight>74</Weight>
    <Filter1>2023W32</Filter1>
    <Filter2 />
    <Filter3 />
    <Filter4 />
    <Filter5 />
    <Filter6 />
    <Filter7 />
    <Filter8 />
    <Filter9 />
    <Quantity>2</Quantity>
    <Rotation>0</Rotation>
    <Parameters />
  </RestPiece>
  <RestPiece>
    <Material>MDF-8</Material>
    <XDimSize>3050</XDimSize>
    <YDimSize>1093</YDimSize>
  </RestPiece>
</DSSRestPieceList>
```

RestPiece nodes:

- MaterialId : ID of the material. If not present, it is derived from the Material- and Color-nodes [optional].
- Material : Material name [optional].
- Color: Material color [optional].
- XDimSize : Length of the sheet (mm, maximum 1 decimal; dot as decimal separator).
- YDimSize : Width of the sheet (mm, maximum 1 decimal; dot as decimal separator).
- ZDimSize : Thickness of the sheet. Default is the thickness of the associated material (mm, maximum 1 decimal; dot as decimal separator) [optional].
- Grain : Grain direction of the sheet: 0=None; 1=Long (parallel to the length); 2=Short (parallel to the width). [optional; default=0]

- Weight : Weight of the sheet (kg, maximum 1 decimal; dot as decimal separator) [optional; if not present, the weight is calculated using the density of the material and dimensions of the sheet].
- Filter1 : Extra attribute for this sheet, e.g. the name of the customer [optional].
- Filter2 : <see Filter1>
- Filter3 : <see Filter1>
- Filter4 : <see Filter1>
- Filter5 : <see Filter1>
- Filter6 : <see Filter1>
- Filter7 : <see Filter1>
- Filter8 : <see Filter1>
- Filter9 : <see Filter1>
- Quantity : Number of sheets [optional; default=1].
- Rotation : Rotation (in degrees) of the sheet(s) on the pick up position, relative to the length of the position. 0=length of the sheet is parallel to the length of the pick up position; 90=width of the sheet is parallel to the length of the pick up position [optional; default=0].
- Parameters : Specific settings for pre- or postprocessing. Contact DSS Automation for details [optional].

Note: To determine the material of the sheet, either *MaterialId* or *Material* (with optional *Color*) is required.

Note: *XDimSize* should always be larger than or equal to *YDimSize*. If not, DSS will swap both values, and change the value for *Grain* accordingly, automatically.

---

## Stocksummary examples

Below a set of files generated by the DSSStockSummaryExport.

Text/csv-file. Number formats follow Windows regional settings. Columns are separated by the COLUMNSEPARATOR character. Texts containing this character are surrounded by double quotes ("). For readability in this manual, the data is presented in a table:

| MATERIAL | COLOR | SUPPLIER | STATUS | HASGRAIN | DENSITY | XDIMSIZE | YDIMSIZE | ZDIMSIZE | GRAIN | QUANTITY |
|---|---|---|---|---|---|---|---|---|---|---|
| IRO065X130 | | UNILIN | Active | False | 900 | 5000 | 130 | 65 | None | 2 |
| MAH065X155 | MAHOGANY | BLOK | Active | True | 660 | 3000 | 155 | 65 | Long | 3 |
| MER GK114 | | | Active | False | 0 | 1690 | 114 | 50 | None | 77 |
| MER GK114 | | | Active | False | 0 | 2249 | 114 | 50 | None | 68 |
| MER XT067 | | | Active | False | 0 | 482 | 67 | 90 | None | 11 |

The same data exported as html-file. Numbers and dates are right-aligned, Booleans are centered:

**DSS Report : TDSSStockSummaryExport**

| MATERIAL | COLOR | SUPPLIER | STATUS | HASGRAIN | DENSITY | XDIMSIZE | YDIMSIZE | ZDIMSIZE | GRAIN | QUANTITY |
|---|---|---|---|---|---|---|---|---|---|---|
| IRO065X130 | | UNILIN | Active | False | 900,0 | 5000,0 | 130,0 | 65,0 | None | 2 |
| MAH065X155 | MAHOGANY | BLOK | Active | True | 660,0 | 3000,0 | 155,0 | 65,0 | Long | 3 |
| MER GK114 | | | Active | False | 0,0 | 1690,0 | 114,0 | 50,0 | None | 77 |
| MER GK114 | | | Active | False | 0,0 | 2249,0 | 114,0 | 50,0 | None | 68 |
| MER XT067 | | | Active | False | 0,0 | 482,0 | 67,0 | 90,0 | None | 11 |

Generated on 17-2-2023 15:42:01 for DSS-VANDEVEN in 00:00.135

The same data exported as xml-file. Floating point values are exported with 1-digit accuracy:

```
<?xml version="1.0" encoding="UTF-8"?>
<!-- DSSSTOCKSUMMARYEXPORT : Generated on 17-2-2023 15:43:04 for DSS-
VANDEVEN -->
<DSSSTOCKSUMMARYEXPORT>
  <ITEM>
    <MATERIAL>IRO065X130</MATERIAL>
    <COLOR/>
    <SUPPLIER>UNILIN</SUPPLIER>
    <STATUS>Active</STATUS>
    <HASGRAIN>False</HASGRAIN>
    <DENSITY>900.0</DENSITY>
    <XDIMSIZE>5000.0</XDIMSIZE>
    <YDIMSIZE>130.0</YDIMSIZE>
    <ZDIMSIZE>65.0</ZDIMSIZE>
    <GRAIN>None</GRAIN>
    <QUANTITY>2</QUANTITY>
  </ITEM>
  <ITEM>
    <MATERIAL>MAH065X155</MATERIAL>
    <COLOR>MAHOGANY</COLOR>
    <SUPPLIER>BLOK</SUPPLIER>
    <STATUS>Active</STATUS>
    <HASGRAIN>True</HASGRAIN>
    <DENSITY>660.0</DENSITY>
    <XDIMSIZE>3000.0</XDIMSIZE>
    <YDIMSIZE>155.0</YDIMSIZE>
    <ZDIMSIZE>65.0</ZDIMSIZE>
    <GRAIN>Long</GRAIN>
    <QUANTITY>3</QUANTITY>
  </ITEM>
  <ITEM>
    <MATERIAL>MER GK114</MATERIAL>
    <COLOR/>
    <SUPPLIER/>
    <STATUS>Active</STATUS>
    <HASGRAIN>False</HASGRAIN>
    <DENSITY>0.0</DENSITY>
    <XDIMSIZE>1690.0</XDIMSIZE>
    <YDIMSIZE>114.0</YDIMSIZE>
    <ZDIMSIZE>50.0</ZDIMSIZE>
    <GRAIN>None</GRAIN>
    <QUANTITY>77</QUANTITY>
```

```
  </ITEM>
  <ITEM>
    <MATERIAL>MER GK114</MATERIAL>
    <COLOR/>
    <SUPPLIER/>
    <STATUS>Active</STATUS>
    <HASGRAIN>False</HASGRAIN>
    <DENSITY>0.0</DENSITY>
    <XDIMSIZE>2249.0</XDIMSIZE>
    <YDIMSIZE>114.0</YDIMSIZE>
    <ZDIMSIZE>50.0</ZDIMSIZE>
    <GRAIN>None</GRAIN>
    <QUANTITY>68</QUANTITY>
  </ITEM>
  <ITEM>
    <MATERIAL>MER XT067</MATERIAL>
    <COLOR/>
    <SUPPLIER/>
    <STATUS>Active</STATUS>
    <HASGRAIN>False</HASGRAIN>
    <DENSITY>0.0</DENSITY>
    <XDIMSIZE>482.0</XDIMSIZE>
    <YDIMSIZE>67.0</YDIMSIZE>
    <ZDIMSIZE>90.0</ZDIMSIZE>
    <GRAIN>None</GRAIN>
    <QUANTITY>11</QUANTITY>
  </ITEM>
</DSSSTOCKSUMMARYEXPORT>
```

The selection of what sheets to list is set by the options INCLUDERESTSHEETS, INCLUDEEMPTYLINES and INCLUDEINOUTPOSITIONS.

---

## Stockmovement-xml example

Below an example of an xml-file generated by DSS after a sheet has been successfully moved.

```
<?xml version="1.0" encoding="UTF-8"?>
<DSSCraneMovementHistory Type="P2P">
  <DateTime>2020-12-20T12:51:05</DateTime>
  <ArticleCode>200641330</ArticleCode>
  <Material Color="" ID="5917">BU013/BELGIAN DESERT 30mm</Material>
  <MaterialName>BU013/BELGIAN DESERT BUV/30</MaterialName>
  <XDimSize>3000</XDimSize>
  <YDimSize>1400</YDimSize>
  <ZDimSize>30</ZDimSize>
  <Grain>None</Grain>
  <Weight>217.8</Weight>
  <Filter1>XPR-2110</Filter1>
  <Filter2>1003</Filter2>
  <Filter3/>
  <SheetBarcode>83KH18U300G</SheetBarcode>
```

```
  <FromStackposition ID="221">ST196</FromStackposition>
  <ToStackposition ID="219">ST194</ToStackposition>
</DSSCraneMovementHistory>
```

Description of the nodes:

- DateTime : Date and time of completion of the movement.
- ArticleCode : Article code of the sheet. For rest pieces, no article code is given.
- Material : Material name, Color and ID as defined by the material library in DSS.
- MaterialName : Material name of the sheet. This may differ from “Material” as it can be modified through an import or by an operator.
- XDimSize : Length of the sheet (mm).
- YDimSize : Width of the sheet (mm).
- ZDimSize : Thickness of the sheet (mm).
- Grain : Grain direction of the sheet. Options: None, Long or Short (or translated values thereof).
- Weight : Weight of the sheet (kg).
- Filter1 : Value of “Filter1”, if any.
- Filter2 : <see Filter1>
- Filter3 : <see Filter1>
- Filter4 : <see Filter1>
- Filter5 : <see Filter1>
- Filter6 : <see Filter1>
- Filter7 : <see Filter1>
- Filter8 : <see Filter1>
- Filter9 : <see Filter1>
- SheetBarcode : Barcode of the sheet.
- FromStackposition : Alias and ID of the former stack position.
- ToStackposition : Alias and ID of the current/new stack position.

The csv-export stores the same data in a tab-separated csv-file, but without the ID-columns.

Using custom-scripts the export can be tailored to the needs and requirements of the customer.

---

## SQL export example

Any valid SQL SELECT statement can be used to generate a report. A few examples below.

**All current Crane Work records (current work queue) :**

```
select
  S.MATERIALNAME, F.ALIAS as FROMSTACK, T.ALIAS as TOSTACK, W.ENTRYDATE
from
  CRANEWORK W
  inner join SHEET S on (S.OBJECTID=W.SHEET)
  inner join STACKPOSITION F on (F.OBJECTID=W.FROMSTACKPOSITION)
```

```
  inner join STACKPOSITION T on (T.OBJECTID=W.TOSTACKPOSITION)
where
  (W.WORKSTATUS<>3)
```

Result:

| MATERIALNAME | FROMSTACK | TOSTACK | ENTRYDATE |
|---|---|---|---|
| SH_RX MULTILENE PE-M_MULTILENE UHMWPE 1000_WHITE_20.0_NA_NA | G2 | LBLZ | 19-3-2021 10:27:57 |
| SH_RX MULTILENE PE-M_MULTILENE UHMWPE 1000_WHITE_20.0_NA_NA | BF2 | ZG2 | 19-3-2021 10:30:53 |
| SH_RX PC STR 40 ZW_PC STR 40 ZW_BLACK_4.0_NO_NA | ZG1 | ZU1 | 19-3-2021 11:39:37 |
| SH_RX MULTILENE PE_MULTILENE HDPE_BLACK_30.0_NA_NA | REST | R-3M-1 | 19-3-2021 11:43:12 |
| SH_EPRAFORM PMMA_PMMA G_CLEAR 100.10000_4.0_NO_NA | LBLZ | BF1 | 19-3-2021 12:09:19 |
| SH_RX PC_PC_CLEAR_10.0_NO_NA | LBLZ | BF1 | 19-3-2021 12:12:19 |
| SH_RX MULTILENE PP-P_MULTILENE PP-P_GREY 7032_100.0_NA_NA | R-1M-1 | LBLZ | 19-3-2021 12:12:37 |
| SH_EPRAFORM PMMA_PMMA G_CLEAR 100.10000_5.0_NO_NA | ZG2 | ZU2 | 18-3-2021 16:20:32 |
| SH_RX MULTILENE PE_MULTILENE HDPE_BLACK_80.0_NA_NA | DIR2 | 103A | 18-3-2021 21:29:27 |
| SH_RX MULTILENE PE_MULTILENE HDPE_BLACK_80.0_NA_NA | DIR2 | <AUTO> | 18-3-2021 21:29:27 |
| SH_RX MULTILENE PE_MULTILENE HDPE_BLACK_6.0_NA_NA | INUIT4 | 012A | 18-3-2021 22:20:24 |
| SH_DILITE_DILITE_WHITE_4.0_NA_NA | 113A | CNC2 | 18-3-2021 22:23:09 |
| SH_RX MULTILENE PE_MULTILENE HDPE_WHITE_10.0_NA_NA | INUIT2 | G4 | 18-3-2021 22:23:55 |
| SH_RX PC_PC_CLEAR_4.0_NO_NA | R-2M-5 | VS3A | 18-3-2021 22:38:10 |
| SH_RX PVC CAW_PVC CAW_DARK GREY 7011_3.0_NO_NA | INUIT1 | <AUTO> | 18-3-2021 23:30:16 |
| SH_RX MULTILENE PE-D_MULTILENE HMWPE 500_BLACK_90.0_NA_NA | 103A | VS3A | 19-3-2021 1:50:08 |

**The contents of stack position “R-1M-3” :**

```
select
  S.ZINDEX, S.MATERIALNAME, round(10*S.XDIMSIZE)/10 as LENGTH,
round(10*S.XDIMSIZE)/10 as WIDTH
from
  SHEET S
  inner join STACKPOSITION P on (P.OBJECTID=S.STACKPOSITION)
where
  (P.ALIAS='R-1M-3') and (S.SHEETSTATUS=0)
order by
  S.ZINDEX desc
```

Result:

| ZINDEX | MATERIALNAME | LENGTH | WIDTH |
|---|---|---|---|
| 12 | SH_PRINT 130_PRINT 130_WHITE_6.0_NO_NA | 1512 | 1512 |
| 11 | SH_RX MULTILENE LDPE-R_MULTILENE LDPE-R_BLACK_30.0_NA_NA | 1500 | 1500 |
| 10 | SH_NYLATRON 703XL_NYLATRON 703XL_PURPLE_10.0_NA_NA | 1538,6 | 1538,6 |
| 9 | SH_RX MULTILENE PE-M_MULTILENE UHMWPE 1000_WHITE_60.0_NA_NA | 1598,6 | 1598,6 |
| 8 | SH_RX MULTILENE PE-D_MULTILENE HMWPE 500_BLACK_70.0_NA_NA | 1273,6 | 1273,6 |
| 7 | SH_RX PVC CAW_PVC CAW_DARK GREY 7011_35.0_NO_NA | 1000 | 1000 |
| 6 | SH_RX MULTILENE PP-X_MULTILENE PP-X_GREY 7032_25.0_NA_NA | 1500 | 1500 |
| 5 | SH_RX EPRAMID G-WAX_EPRAMID G-WAX_GREY_60.0_NA_NA | 1195 | 1195 |
| 4 | SH_RX MULTILENE PE-DR_MULTILENE HMWPE 500REG_BLACK_45.0_NA_NA | 1030,8 | 1030,8 |
| 3 | SH_EPRAFORM PMMA_PMMA G_CLEAR 100.10000_120.0_NO_NA | 1113,6 | 1113,6 |
| 2 | SH_RX EPRADUR P-B_EPRADUR P-B_DARK GREY 7011_60.0_NO_NA | 1095 | 1095 |
| 1 | SH_RX MULTILENE PE-MR_MULTILENE UHMWPE 1000REG_GREEN_10.0_NA_NA | 1490 | 1490 |

## Configuring database link

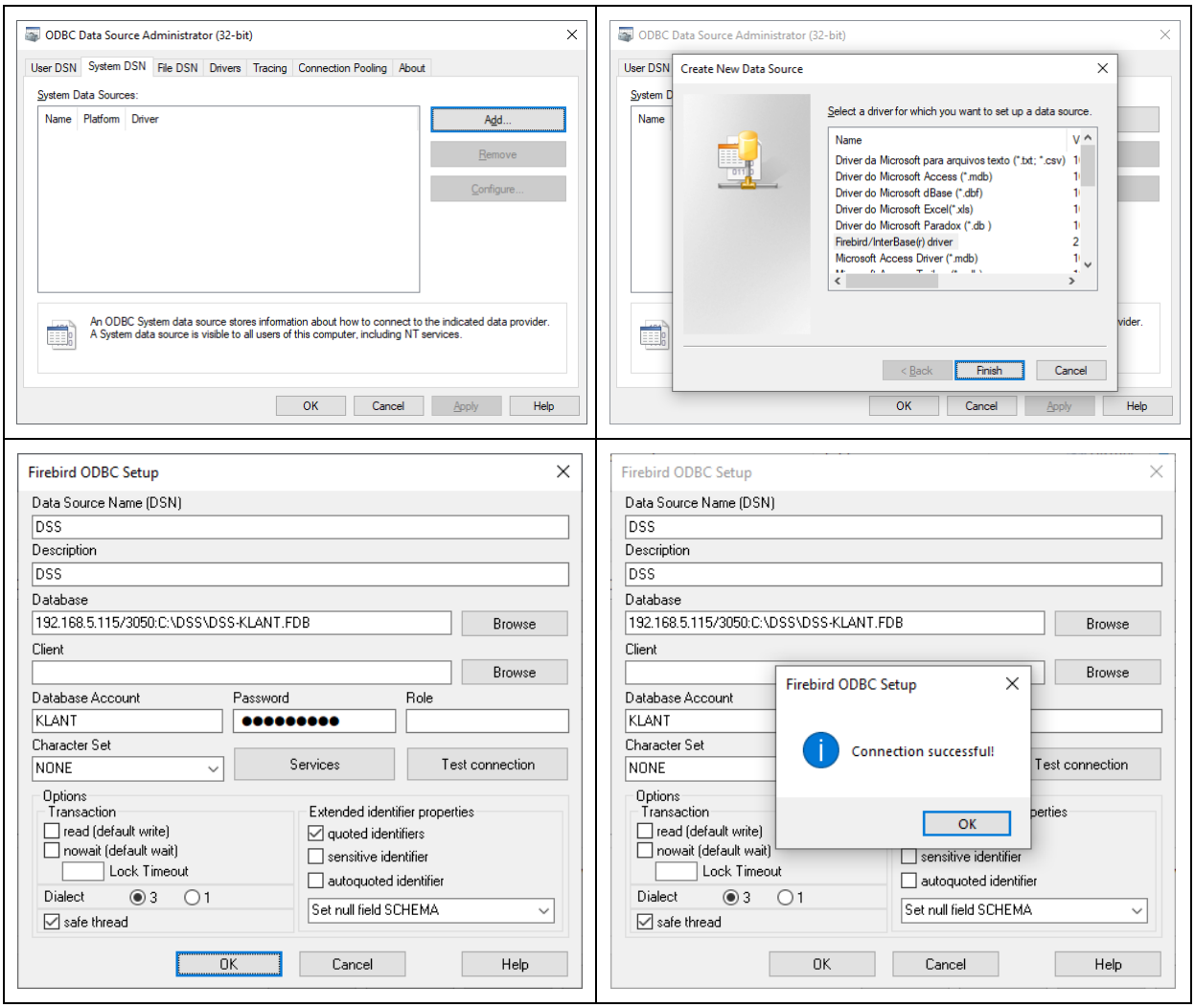
The current versions of the Firebird ODBC driver are:

`Firebird_ODBC_2.0.4.155_Win32.exe` (for 32-bit Windows)
`Firebird_ODBC_2.0.4.155_x64.exe` (for 64-bit Windows)

Run the proper installer. When done:

On Win-32bit : Copy `gds32.dll` from `gds32-32bit.zip` to `C:\Windows\System32\`.
On Win-64bit : Copy `gds32.dll` from `gds32-64bit.zip` to `C:\Windows\SysWOW64\`.

Next, open the **ODBC Data Source** app add a new System-DSN:




Enter the following details:

- Database: {ip-address of the computer de database is running on}`/3050:C:\DSS\DSS-`{customer_name}`.FDB`
- Database Account: {customer_name}
- Password: {customer_name}

And click the [Test connection] button to verify all is functioning properly. In the remote application, run the query `select * from XX_{customer_name}_STOCK` to retrieve the contents of the view.

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